Promo Finder: picking microdrama promos from the screenplay

Kuku AI Generalist assignment · Problem 1 · ran on all 50 episodes of *I Married a Broken Soldier. He was a Duke* (50 episodes, 103 min, 3,727 timestamped beats)

Prototype: promo-finder-ahmed2062.streamlit.app · **Code + raw outputs:** github.com/Ahmed2062/promo-finder · full unredacted outputs shared privately as a zip

1. Why this problem, and how I scoped it

Promo selection is a weekly job that blocks every launch. It has a clear spec (20–60 s, hook in 3 s, cut on a cliffhanger, stands alone), and you can check it from the script alone, so an automated pass can do the first 90% and an editor the last 10%. I scoped it to **finding and ranking cuts**: no editing, music or overlays, as the brief says.

Assumptions.

- Script timestamps match the final cut closely enough to hand an editor a starting in/out point. They won't be frame-accurate, so the editor still trims.
- A clicked ad lands the viewer on episode 1, so promos from early episodes are worth more. The ranking gives a small bonus for that.
- Promos may spoil early twists but not the series' central reveal (the husband is the Duke).
- A promo can join 2–3 moments, from one episode or several.

2. How it works: the LLM points, code measures

- **Parse (code).** A forgiving parser turns each script into beats with stable ids (`E12.B041 [01:03.10-01:05.40] CLARISSA (Cold): …`). From these it computes **safe cut points**, where nobody is mid-sentence.
- **Shortlist (code, free, 6 s).** Keyword and structure scores per beat: hook (violence, humiliation, reveals, emotion tags), cliffhanger (questions, threats, ultimatums, interruptions) and resolution (hugs, smiles, resolving music). It scores every 20–60 s window and builds stitched hook + payoff combos. This is the LLM's hint list and also the no-key fallback.
- **Pick (LLM, 1 call per episode).** The model gets the story so far (loglines only), the episode as numbered beats and the shortlist. It answers with **beat ids, never timestamps**, plus hook, cliffhanger, why it works and 0–10 scores. One more call builds cross-episode promos.
- **Repair + check (code).** Cuts snap to safe points. A promo over 60 s becomes hook (≈8 s) + payoff, as an editor would cut it; one under 20 s starts earlier. Hard rules: length, clean cuts, at least 3 dialogue lines, no tiny segments. The keyword hook and cliffhanger checks are hard for heuristic picks but only **review flags** for LLM picks, because in my audit the keyword checker was usually the one that was wrong.
- **Rank (code + 1 LLM call).** Score = 0.32 hook + 0.30 cliff + 0.20 standalone + 0.08 intensity + 0.10 funnel – penalties. Hook, cliff and standalone each blend 70% model score with 30% heuristic score. The LLM then sets the posting order from the top 25, gives a reason for each slot and flags weak or duplicate picks.
- **Blind viewer (1 LLM call, QA only).** A separate prompt sees only the clip's lines, with no show context, and rates stop / clear / next. A clarity score of 3 or below flags the promo for a human.

Human in the loop: the editor gets `report.html`, a ranked list showing each cut on a timeline with its checks, review flags and transcript. They approve or adjust instead of watching 100 minutes. **Model swap:** `--provider anthropic|gemini|groq|openrouter|ollama|openai|manual|none` plus `--model`. Every prompt and raw reply is saved and cached, so reruns are free and reproducible.

3. Output: what I would post first

The model returned picks for 46 of 50 episodes and skipped four with no hook (E06, E08, E17, E40). That gave 75 promos after de-duplication, of which 15 are ranked. Open with the early, unmistakable premise (jilted bride and a scarred stranger's proposal), because viewers land on episode 1 and E1-E3 promos pay off fastest. Then rotate the angles (cruel sister, protector romance, ring mystery, villain menace, slow-burn romance, revenge tease) so that no two wedding-scene promos run back to back. Late-episode material is held to teases that hint at the Duke reveal without spoiling it.

#	Segments (episode, mm:ss)	Hook → cliffhanger	Why post it here
1	E3 00:58.3–01:25.4 + E3 01:59.2–02:25.7 (54s)	Matriarch screams 'Guards! Remove her!' and guards drag the bride from the church as she begs 'Please! Someone → *The scarred stranger kneels in the aisle with a ring - 'Lady Juliana... accept my hand' - and we cut*	Strongest hook and cliff of the set: a bride dragged from the church, then a scarred stranger kneels with a ring. Pays off within minutes of landing on episode 1.
2	E12 00:44.0–01:27.2 (43s)	Clarissa, smiling: 'I want to see you crawl, sister.' → *Juliana accepts the ordeal, 'Tell me where the garden is', and we freeze on her face before she reac*	Switches to the cruel-sister angle with a chilling 'I want to see you crawl' hook. It ends on her accepting the ordeal, so the viewer needs to see what happens next.
3	E15 00:44.2–01:26.0 (42s)	A thug snarls 'Get out of the way!', Juliana says 'No', and he shoves her hard onto the wet stone → *Alexander stands between her and three armed thugs: 'What? You want to fight us, veteran?' and we cu*	Protector romance: she is shoved into the mud and the veteran husband steps in against three thugs. Easy for a stranger to follow, and still early in the show.
4	E1 00:33.8–00:50.0 + E3 00:58.3–01:04.5 + E3 02:14.8–02:25.7 (33s)	A bride in her mother's gown whispers 'Gerald? Why are you holding her hand?' as her groom stands at the altar → *The whole congregation gasps as a rugged scarred stranger kneels before the dragged-out bride; she s*	A tight 33s version of the jilted-at-the-altar premise for viewers who missed slot 1. It hooks on the betrayal from E1 and cuts on the stranger kneeling.
5	E26 00:56.0–01:06.7 + E26 01:26.5–02:00.8 (45s)	A dying duchess whispers 'Rupert and Catherine want me dead. They are poisoning me in my own bed, aren't they? → *The Duchess grips Juliana's hand - 'Do not move! Where did you get this ring? Speak to me!' - cut be*	Adds mystery: a dying Duchess recognizes the ring and demands to know where it came from. It plants the hidden-Duke question without answering it.

6	E2 00:00.0–00:12.0 + E2 00:47.8–01:16.4 (41s)	Groom at the altar tells his bride 'I cannot marry you, Juliana. You are far too plain...' → *Told to 'go home quietly with some dignity', Juliana turns back with a steely smile: 'Dignity?' - we*	Humiliation into defiance: 'too plain to marry' turns into a steely 'Dignity?' comeback. Very clear archetype from the opening episodes.

4. How I know it's good, and where it picks badly

- **Rule checks:** 89% of LLM promos pass every hard rule. 0 have bad lengths or mid-line cuts; that is guaranteed, because code owns the timestamps. 28 needed a sub-second snap to a clean cut. 0 LLM items were rejected for invalid beat ids.
- **Model vs. heuristics:** 81% of LLM picks overlap one of the heuristic's top-2 windows for that episode, so the two methods agree on *where* the moments are. They disagree on hook and cliff quality: 18 hook and 13 cliffhanger flags. I read 10 of them by hand: in about 7 the keywords were wrong (strong lines like "Take her." and "Your Grace." contain no keywords), and in about 3 the flag was right (E27 opens on a flat "It is an antidote.").
- **Blind viewer:** stop 5.6/10, clear 4.8/10, next 5.5/10. Only 20% would click. That is harsh, but it is the most useful signal here, and it disagrees with the ranker where it should.

Where it fails

- **Stitched cross-episode promos lose the story.** X-02 (mud + ledger → "he has the late Duke's scar") scored 7/10 standalone from the picking model but 2/10 clarity from the blind viewer. X-01's stranger proposes "out of nowhere" because the cut drops his entrance. Fix: require a character-intro beat in every stitched promo.
- **Romance and quiet beats don't read as text.** E10's near-kiss scored 3/10 stop. The tool can't see acting, chemistry or visuals, which are what sell these. It needs a video pass (frames + audio) to judge them.
- **The script names the same character several ways** (BEARDED STRANGER / MYSTERIOUS BEARDED MAN / ALEXANDER). That confuses standalone judgement; a character-alias pass would fix it.
- **The heuristics trust the script's own notes** ("cliffhanger swell"). They counted a "resolving chord" as a cliffhanger until I added resolution words, and keyword lists will always miss understated lines.
- **Late episodes spoil the main reveal.** The ranker flagged E42 and E43 for this, and the funnel bonus pushes them down; 5 of the 15 ranked promos come from E1–E10.

5. Scale and cost

One show is about 258K input and 17K output tokens over 53 calls. At list prices that's ≈$0.69 on Claude Sonnet 5.5, ≈$1.37 on Opus 5.5 and ≈$0.03 on Haiku 5.5; free tiers (Gemini, Groq) or Ollama cost $0. The code stages take under 15 s. Per-episode calls run in parallel or through a batch API, and the story-so-far prefix can be cached. A new show every week is a one-command job plus about 20 minutes of editor review instead of 2+ hours of watching. Next steps: align the script to the actual video (ASR) for frame-accurate cuts, and learn the ranking weights from real promo CTR/CPI once posts are live.

6. AI tools, and one place they got it wrong

I used Claude Code (Claude Opus 5.5) to design and write the code, and Claude as the LLM inside the pipeline. I had no free API key on this machine, so I ran the LLM stages through the tool's `manual` provider: each saved prompt was answered verbatim by a separate Claude agent that saw only that prompt. All raw prompts and replies are in `outputs/llm/llm/`, and any API provider reruns them with one flag. **Where AI got it wrong:** the picking model rated the stitched X-02 promo 7/10 for standing alone and the ranker put it 8th, but a context-free reader found it incoherent. Also, the heuristic Claude first wrote for me treated every episode ending as a cliffhanger, even calm, resolved ones, until the baseline run showed a "resolving chord" ranked 9th.